Sample QUESTION PAPER

Total Marks: 50
Time: 2 Hours

General Instructions:

- The question paper is divided into three sections: Section A, Section B, and Section C.
- **Section A** contains 14 Multiple Choice Questions of 1 mark each.
- **Section B** contains 10 Short Answer Questions of 2 marks each.
- **Section C** contains 4 Long Answer / Application Questions of 4 marks each.
- Read all scenarios carefully before answering.

SECTION A – MULTIPLE CHOICE QUESTIONS (1 Mark Each)

Q1. An automated smart car suddenly slams on its brakes on an empty highway because its camera system mistook a dark shadow on the road for a large rock. In the context of model evaluation, which type of error did the car's AI commit?
A) True Negative
B) True Positive
C) False Positive
D) False Negative

Q2. A global news platform uses an algorithm that reads articles in Spanish and automatically translates them into English text, while simultaneously categorizing them under "Politics" or "Sports". Which two AI domains are primarily functioning together in this application?
A) Computer Vision and Data Sciences
B) Natural Language Processing (NLP) and Data Sciences
C) Natural Language Processing (NLP) and Computer Vision

D) Rule-Based Systems and Deep Learning

Q3. A teacher is tracking the attendance of students over a month. Most students have 100% attendance, but two students were severely ill and missed almost the entire month, pulling the average drastically down. If this data were plotted on a graph, what type of distribution would it represent?
A) Perfect Normal Distribution
B) Left-skewed (Negative Skew) Distribution
C) Right-skewed (Positive Skew) Distribution
D) Unbalanced Symmetrical Distribution

Q4. An indie video game developer with no programming background wants to create an AI opponent that learns a player's combat style through a webcam. Which specific web-based tool from your syllabus allows them to train this model purely through a visual interface without writing code?
A) Microsoft Azure Java Interface
B) Teachable Machine
C) Orange Data Mining Text Analyzer
D) High-Code Python Environment

Q5. A smart fitness watch is programmed with the following logic: "If the user's heart rate exceeds 160 BPM for more than 5 minutes, trigger a loud alarm." What kind of system is this?
A) Unsupervised Machine Learning
B) Deep Learning Convolutional Network
C) Learning-Based Approach
D) Rule-Based Approach

Q6. A supermarket manager wants to analyze a spreadsheet of daily sales. Some cashiers forgot to log the sale amounts for a few specific transactions, leaving empty cells. The manager calculates the average sale value and fills those blank cells with that average. What is this specific data preparation technique called?
A) Imputation
B) Dimensionality Reduction
C) Association
D) Clustering

Q7. A voice-activated virtual assistant is deployed in a rural town. However, it completely fails to understand the commands of the local residents because it was trained entirely on audio clips from city residents. Which core ethical concern does this AI system violate?
A) Accountability
B) Fairness and Bias
C) Data Privacy
D) Transparency

Q8. A meteorologist builds a weather prediction model. It performs excellently when evaluating the "Training Subset" of data from 2010 to 2020. However, it performs terribly when tested on the "Testing Subset" from 2021 to 2023. What technical term best describes this phenomenon?
A) Underfitting
B) Perfect Fit
C) Overfitting
D) High Bias

Q9. A grocery store AI automatically notices that customers who buy diapers on Friday evenings almost always buy baby wipes and milk as well. It doesn't classify these items, but merely finds the hidden relationships between them. Which machine learning technique is this?
A) Supervised Regression
B) Supervised Classification
C) Reinforcement Learning
D) Unsupervised Association Rule Learning

Q10. When building a workflow in Orange Data Mining, a student wants to see a visual table comparing the model's predicted labels against the actual labels to see where the AI got confused. Which specific widget must they drag onto the canvas?
A) Confusion Matrix Widget
B) Data Sampler Widget
C) Impute Widget
D) Scatter Plot Widget

Q11. A developer creates a highly sensitive AI tool for early earthquake detection. If an earthquake is actually coming, failing to trigger the alarm could cost thousands of lives. Therefore, the developer must design the system to absolutely minimize which metric?
A) True Positives
B) False Positives
C) False Negatives
D) True Negatives

Q12. An AI plays a maze-solving game. It explores random paths. If it hits a dead end, it receives a "-10" penalty. If it finds the exit, it gets a "+100" reward. After 1,000 attempts, it navigates the maze flawlessly. Which learning model is this?
A) Unsupervised Learning
B) Reinforcement Learning
C) Natural Language Processing
D) Regression

Q13. In a standard medical dataset of 1,000 patients, 990 are perfectly healthy and only 10 have a rare disease. An AI model is built to detect this disease. Why is the "Accuracy" metric considered dangerous to use on this specific dataset?
A) It cannot be calculated mathematically.
B) The dataset is highly unbalanced, so the model could just guess "healthy" every time and still get 99% accuracy while missing every sick patient.
C) Accuracy ignores True Negatives completely.
D) Accuracy is only used for Unsupervised Learning.

Q14. During the AI Project Cycle, a team identifies the key stakeholders, documents the context of the issue, maps out where the issue occurs, and defines why it is valuable to solve. Which stage is the team currently executing?
A) Data Acquisition
B) Evaluation
C) Problem Scoping
D) Modelling

SECTION B: SHORT ANSWER QUESTIONS (2 Marks Each)

Q15. An AI-powered email app offers a feature to automatically sort incoming

messages into "Urgent" or "Promotional" folders. Distinguish between how this would be achieved using a *Rule-Based Approach* versus a *Learning-Based Approach*.

Q16. A school is testing a new AI grading system. Out of 100 student essays, it perfectly scores 90 of them correctly, but flags 10 highly original essays as "Plagiarized" when they were entirely written by the students.
Identify the type of error (TP, TN, FP, or FN) the AI made on those 10 essays and explain the negative real-world impact of this error on the students.

Q17. A retail brand creates an AI system to analyze customer feedback surveys. While reviewing the workflow in Orange Data Mining, the lead analyst realizes they skipped the "Train-Test Split" process entirely and tested the model on the exact same data they trained it on. Explain why this mistake renders their evaluation results invalid.

Q18. A bank rejects a young entrepreneur's small business loan using an automated AI system. When the entrepreneur asks for the specific reason, the bank manager says, "We don't know, the neural network is a black box and just said no." Identify the AI ethical principle being violated here and briefly explain why it is crucial in this scenario.

Q19. A data scientist analyzes the monthly playtime of users on a mobile game. They calculate both the Mean (average) playtime and the Standard Deviation. Why is it necessary to look at the Standard Deviation instead of just relying solely on the Mean?

Q20. A logistics company trains a Convolutional Neural Network (CNN) to automatically recognize and sort different types of fruit on a fast-moving conveyor belt. Briefly explain the specific function of the *Input Layer* and the *Output Layer* in this neural network.

Q21. An AI system is designed to predict daily stock market prices. After a week of testing, it becomes obvious that the model is "Underfitting." Define what Underfitting means in terms of the model's complexity and its performance on both training and testing data.

Q22. A hospital administrator with no coding background needs to build an AI workflow to analyze patient recovery times. They decide to use a No-Code AI

platform. State two distinct risks or disadvantages they might face by choosing a No-Code platform for highly sensitive healthcare data.

Q23. Distinguish between the functions of "Visualize Widgets" and "Evaluate Widgets" in the Orange Data Mining platform. Provide one example of a widget belonging to each category.

Q24. In a criminal justice AI system (where a person is innocent until proven guilty), the algorithm is designed to flag individuals who are highly likely to re-offend. Explain why developers must prioritize the *Precision* metric over *Recall* in this high-stakes scenario.

SECTION C: LONG ANSWER QUESTIONS (4 Marks Each)

Q25. A city traffic department deploys an AI camera system at a busy intersection to automatically issue tickets to drivers who run red lights. Over one week, the system processes 200 vehicles and generates the following Confusion Matrix:

- **True Positives (TP):** 60 (Actual red-light runners correctly ticketed)
- **True Negatives (TN):** 120 (Law-abiding drivers correctly ignored)
- **False Positives (FP):** 20 (Law-abiding drivers incorrectly issued a ticket)
- **False Negatives (FN):** 0 (No red-light runners were missed)

a) Mathematically calculate the total Accuracy of this AI system as a percentage. *(2 marks)*
b) Mathematically calculate the Precision of this AI system. *(1 mark)*
c) Based on the False Positives generated, describe the real-world consequence this system will have on the city's citizens if deployed without human review. *(1 mark)*

Q26. A regional agricultural board wants to build an AI app that helps local farmers detect early signs of a specific fungal disease on wheat crops just by taking a photo on their smartphones.
Apply the first four stages of the **AI Project Cycle** to this specific scenario. For each of

the following stages, describe one distinct, practical action the development team must execute:

1. Problem Scoping
2. Data Acquisition
3. Data Exploration
4. Modelling

Q27. An e-commerce giant wants to launch a new product recommendation engine. They are debating between two different Artificial Intelligence architectures:

- **Architecture A:** Uses standard Artificial Neural Networks (ANN) to process lists of text-based purchase histories.
- **Architecture B:** Uses Convolutional Neural Networks (CNN) to visually scan and analyze the actual product photographs customers spend the most time hovering over.

a) Briefly explain the biological inspiration behind both of these architectures (how do they mimic the human body?). *(1 mark)*
b) Contrast these two architectures by explaining why Architecture B (CNN) is vastly superior to Architecture A (ANN) specifically for analyzing the product photographs. *(2 marks)*
c) Identify the specific AI domain associated with Architecture B's task. *(1 mark)*

Q28. An educational software company is developing an AI tutor. They feed the AI millions of past student math tests.

- **Scenario X:** The AI looks at tests that have already been graded by human teachers (marked as "Pass" or "Fail") and learns the patterns of a failing test to grade future tests.

- **Scenario Y:** The AI looks at a massive pile of ungraded, unlabeled tests and automatically groups students who make similar types of fraction errors into distinct clusters so teachers can help them in groups.

a) Identify the specific category of machine learning used in Scenario X and explain how it utilizes data. *(2 marks)*
b) Identify the specific category of machine learning used in Scenario Y and explain how it utilizes data. *(2 marks)*